

Constructions (Set A) — MCQ Worksheet

Mathematics • Chapter 11 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

- Q1.** To divide a line segment in a given ratio $m:n$ we mark on a ray the total of:
- (A) m points (B) n points
(C) $(m + n)$ points (D) $m \times n$ points
- Q2.** To divide a line segment AB in ratio 3:5 we mark a total of how many points on the ray?
- (A) 3 (B) 5
(C) 8 (D) 15
- Q3.** While dividing a line segment, we draw a ray from one end making:
- (A) Right angle (B) Acute angle
(C) Obtuse angle (D) Straight angle
- Q4.** While constructing similar triangle with a scale factor $\frac{3}{5}$, we mark on the ray:
- (A) 3 points (B) 5 points
(C) 8 points (D) 2 points
- Q5.** When constructing similar triangle with scale factor $\frac{m}{n}$ where $m > n$, we mark how many points?
- (A) $m + n$ (B) m
(C) n (D) Greater of m, n
- Q6.** To construct a triangle similar to a given triangle with scale factor $\frac{7}{5}$, the new triangle will be:
- (A) Larger than given (B) Smaller than given
(C) Same size (D) Inverted
- Q7.** To construct a triangle similar to given triangle with scale factor $\frac{3}{4}$ means:
- (A) New is larger (B) New is smaller
(C) Same size (D) No relation
- Q8.** To construct tangents to a circle from external point P, we first draw:
- (A) A line through centre (B) A perpendicular bisector
(C) Circle with diameter OP (D) Random arc
- Q9.** In tangent construction from an external point, the perpendicular bisector of OP is drawn to find:
- (A) Tangent length (B) Midpoint M of OP, then circle with M as centre
(C) Random (D) None
- Q10.** In tangent from external point construction, the auxiliary circle has centre at:
- (A) O (B) P
(C) Midpoint of OP (D) On the original circle
- Q11.** The number of tangents constructed from an external point is:
- (A) 1 (B) 2
(C) 3 (D) 4
- Q12.** In a construction, an arc is drawn using:
- (A) Ruler (B) Set square
(C) Compasses (D) Protractor
- Q13.** Justification of constructions involves:
- (A) Stating principles like BPT, similar triangles (B) Listing materials
(C) Drawing freehand (D) Random checking
- Q14.** To construct a triangle ABC similar to a given triangle with scale factor k , the corresponding sides will be in ratio:
- (A) k (B) $k:1$
(C) $1:k$ (D) k^2

- Q15.** While dividing a segment in ratio 2:3, the points marked are equally spaced. This uses:
(A) BPT (B) Pythagoras
(C) Trigonometry (D) Distance formula
- Q16.** The construction of an angle bisector uses:
(A) Ruler only (B) Compass and ruler
(C) Protractor only (D) Set square
- Q17.** The locus of points equidistant from two given points is the:
(A) Angle bisector (B) Perpendicular bisector
(C) Median (D) Altitude
- Q18.** To draw a perpendicular to a line at a point on the line, we use:
(A) Compass arcs above and below (B) Protractor
(C) Set square only (D) Free-hand
- Q19.** Constructing a triangle requires at least ____ measurements:
(A) 1 (B) 2
(C) 3 (D) 4
- Q20.** To construct a 60° angle using compass we draw an arc cutting the line at A and from there a same-radius arc to fix:
(A) A point B; $\angle$ at start = 60° (B) A right angle
(C) A straight line (D) A circle
- Q21.** To bisect a 60° angle, the resulting angle is:
(A) 30° (B) 45°
(C) 15° (D) 60°
- Q22.** While constructing a tangent to a circle from a point on the circle, we need:
(A) Perpendicular at that point to the radius (B) Ruler
(C) Bisector (D) Two arcs
- Q23.** Two tangents drawn from an external point are:
(A) Of unequal length (B) Of equal length
(C) Always parallel (D) Always perpendicular
- Q24.** To construct a triangle similar with scale factor 1, the triangle is:
(A) Larger (B) Smaller
(C) Congruent (D) Mirror image
- Q25.** In dividing line segment AB in the ratio m:n internally, the dividing point lies:
(A) Beyond B (B) Between A and B
(C) Beyond A (D) On a perpendicular